7+1 repository after the clean-up of 28 October 2026: symbols and terms

A dictionary of every symbol and term of the document on the state of the repository at commit 102f399 and of the spreadsheet 7plus1_status_2026-10-28.xlsx, with the defining equations

Repository: github.com/once-ere/7plus1 — prepared for Nash
Commit described: 102f399 of 28 October 2026, 13:15 PDT (branch main)
Document generated by tools/make_docs.py on 2026-10-28

Abstract

For each symbol or term this dictionary gives its definition and its meaning, grouped by topic: the 7+1 spacetime, the repository and its commits, the documents and TeX's measure of a loose line, the kinds of rewritten paragraphs, the checks, the spreadsheet and its functions, and every symbol of the formulas of the rewritten paragraphs. Longer definitions follow each table as numbered displayed equations. Section 10 lists the file and program names, section 11 the symbols that look alike. Every equation needed to define a symbol is written in this document.

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1. Conventions

- Coordinates $X = (x_0, x_1, \ldots, x_7)$: x_0 is time, the direction of the octonion unit $e_0 = 1$; x_1, x_2, x_3 ordinary space; $x_4, \ldots, x_7$ the four auxiliary space-like dimensions. Indices: $A, B = 0, \ldots, 7$; $\mu, \nu, \rho, \sigma, \lambda = 0, \ldots, 7$ for coordinate indices; $i, j = 1, \ldots, 7$ for the space directions; repeated indices are summed unless stated otherwise.
- Units: $\hbar = c = 1$; masses and energies in GeV.
- Matrices: 1_n is the $n \times n$ identity, M^{T} the transpose, $M^\dagger$ the conjugate transpose, M^* the complex conjugate (section 9.14 writes them out); δ_{ij} is the Kronecker delta (1 for $i = j$, 0 otherwise).
- Christoffel symbols are written $\{{}^{\lambda}_{\mu\nu}\}$, never Γ: in this project Γ always denotes a Dirac matrix.

The signature: $\{7, 1, 0\}$ and $\mathrm{diag}(+1, -1, \ldots, -1)$. The 7+1 spacetime has one time-like dimension (x_0, the direction of the octonion unit $e_0 = 1$) and seven space-like dimensions (x_1, x_2, x_3 ordinary space, $x_4, \ldots, x_7$ auxiliary space). Two different objects describe this, and neither may be replaced by the other. (i) The tensor package xAct, used in section 12 of 7+1.nb exactly as in the author's notebook Einstein-Lovelock-7+1-Nash.nb, declares the metric with `DefMetric[{7, 1, 0}, g71[-a, -b], CD, …]`: the list $\{7, 1, 0\}$ declares seven space-like dimensions, one time-like dimension and no degenerate dimension — the numbers of entries $+1$ and -1 of $\eta_{71} = \mathrm{diag}(-1, +1, \ldots, +1)$ (`η71`), the same spacetime written in the sign convention of 4+4.nb (space-like = $+1$, time-like = -1). (ii) The matrix $\eta = -\eta_{71} = \mathrm{diag}(+1, -1, \ldots, -1)$ (`η17`) is simply the squares of the octonion basis: $e_0^2 = +1$ for the unit $e_0 = 1$ (time) and $e_i^2 = -1$ for the seven imaginary units $e_1, \ldots, e_7$ (space); more generally it is the real part of the product of two basis units, and it gives the quadratic form of the octonions:

$$\eta_{AB} = \mathrm{Re}(e_A e_B), \qquad e_0^2 = +1, \qquad e_i^2 = -1 \quad (i = 1, \ldots, 7),$$
$$\mathrm{Re}(x\,x) = x_0^2 - x_1^2 - \cdots - x_7^2 = \eta_{AB}\, x^A x^B \qquad (x = x^A e_A \in \mathbb{O}).$$

η gives the signs of the Clifford relation $\Gamma_A \Gamma_B + \Gamma_B \Gamma_A = 2\,\eta_{AB}\, 1_{16}$ of the real 16×16 Dirac matrices: time squares to $+1_{16}$ and each of the seven space directions to -1_{16}, exactly like the octonion units (in the notation $Cl(p, q)$, with p generators squaring to $+1$ and q to -1, the algebra $Cl(1, 7) \cong \mathbb{R}(16)$). It also gives the signs of the metric components: $g_{00} > 0$ along the time-like direction, negative squares along the space-like directions. Every component computed by 7+1.nb follows from the component matrix of the metric; the declared list fixes for xAct the sign of the determinant of the metric, `SignDetOfMetric[g71]` $= -1$, which is the same in both sign conventions, since $\det \eta = \det \eta_{71} = -1$. 7+1.nb checks `SignatureOfMetric[g71]` $= \{7, 1, 0\}$, `SignDetOfMetric[g71]` $= -1$, $\mathrm{Re}(e_A e_B) = \eta_{AB}$ and $\mathrm{Re}(x\,x) = \eta_{AB} x^A x^B$.

2. The repository and its commits

symbol or term	definition	meaning
repository	github.com/once-ere/7plus1	the Git repository of the 7+1 project, on GitHub
branch `main`	the line of commits that is published	the commits of the repository
commit	a recorded state of all files	one step of the history; its parent is the commit before
commit name (`102f399`)	the first 7 hexadecimal digits of the 40-digit SHA-1 object name	short name; the full name of 102f399 is `102f399d53720ba9b6bc05c6dbf8d70c68c9ac30`
commit described	102f399	the state of the repository that the spreadsheet records
F_c, A_c, R_c	(2.1)	files changed, lines added, lines removed by the commit c
F, A, R	(2.2)	the totals of the four commits
file change	one file changed by one commit	a file changed by two commits is counted twice
binary file	a file that Git does not compare line by line (a PDF)	counts as changed, adds and removes no lines
fresh clone	`git clone https://github.com/once-ere/7plus1.git` into an empty folder	a copy that contains only what is published
tracked file	a file listed by `git ls-files`	a file that is part of the repository

symbol or term	definition	meaning
`.gitignore`	a file of patterns	files that Git does not track; this one is not published itself
`.git/info/exclude`	a file of patterns in one working folder	files that Git does not track there; never published
private file	a file of the author in the working folder, not tracked	not on GitHub
SHA-256	a 256-bit cryptographic hash, written as 64 hexadecimal digits	the fingerprint of the contents of a file
PDT	UTC−7	Pacific Daylight Time, the time zone of the author's computer

$$(F_c,\ A_c,\ R_c) = \text{the numbers of } \texttt{git show --shortstat} \text{ for the commit } c, \tag{2.1}$$

$$F = \sum_c F_c, \qquad A = \sum_c A_c, \qquad R = \sum_c R_c \qquad (c = \texttt{0654dcd},\ \texttt{54862f0},\ \texttt{93c500f},\ \texttt{102f399}). \tag{2.2}$$

For the four commits of 28 October 2026: $F = 122+12+44+28 = 206$, $A = 24996+66+210+374 = 25646$, $R = 4890+19+93+355 = 5357$.

3. Documents and typesetting

symbol or term	definition	meaning
document	one documentation file, as `.md`, `.tex` and `.pdf`	Markdown for GitHub, LaTeX source, compiled PDF
kind	Main, Symbols or Provenance	a document, its symbol dictionary, its provenance record
covers	sections of 7+1.nb	what the document describes
pages	the number reported by `pdfinfo`	the length of the PDF
w	the width of a line	the box that TeX fills
w_0	natural width	the width of the material of the line with every space at its natural size
s, z	total stretchability, total shrinkability	by how much the spaces of the line may grow or shrink
t	$w - w_0$ (when positive)	the stretch that the line needs
b	(3.1)	badness: how much a line is stretched, from 0 (natural) to 10000 (infinitely bad)
`\hbadness`	1000 (set by LaTeX)	the badness above which TeX reports a box
`\hfuzz`	0.1 pt (set by LaTeX)	the excess width that TeX tolerates silently
loose line	(3.2); the message `Underfull \hbox`	a line whose spaces stretch too much
overflowing line	(3.3); the message `Overfull \hbox`	a line wider than the line width
LaTeX error	a line of the log that begins with "!"	a fault in the LaTeX source
layout problem	a loose or overflowing line, an error, or a missing PDF	what the rebuild must not show
`latexmk -xelatex`	the program that runs XeLaTeX until the cross-references settle	compiles each `.tex` file inside build/docs
`pdftotext`	the text of a PDF	used to compare two PDFs
first build	the first full build of the documents on 28 October 2026	before commit 0654dcd: 40 loose lines
changed by a commit	the commit changed the `.md`, `.tex` or `.pdf` file of the document	

$$b = \min\left(10000,\ 100\Big(\frac{t}{s}\Big)^3\right) \quad (s > 0), \qquad b = 10000 \quad (s = 0,\ t > 0), \tag{3.1}$$

$$\text{loose} \iff b > 1000 \iff \frac{t}{s} > 10^{1/3} \approx 2.154, \tag{3.2}$$

$$\text{overflowing} \iff w_0 - z - w > 0.1 \text{ pt}. \tag{3.3}$$

TeX rounds b to an integer. The number of loose (overflowing) lines of a document is the number of messages `Underfull \hbox` (`Overfull \hbox`) in the log of its last LaTeX run; each message is one box, a line of a paragraph or of a table cell.

4. The kinds of rewritten paragraphs

"How"	**definition**	**example**
Line breaking	a change of the typesetting rules of all documents	code may break after `/`, `_` and `\`, a long hexadecimal string every 8 characters; headings are set ragged right (not stretched to the full width)
Display	a long inline formula moved into a displayed equation	the definitions of Θ^K, σ_r and Θ_K (§9.1)
Code block	a long piece of code moved into a block of its own	the xAct definitions (§9.3)
Reworded	the sentence rewritten so that TeX can break its lines evenly	the electroweak scale (§9.13)
Correction	a correction of content	the check counts 50 and 38 (before, 49 and 37)

A paragraph that is rewritten keeps its content. "Paragraphs rewritten" counts the rows of the kinds Display, Code block and Reworded.

5. The checks

symbol or term	**definition**	**meaning**
check program	one of the twelve Python programs of tools/ (section 10)	recomputes results independently, or parses the notebook
exit code	the status with which a program ends	0 means success
ℓ	a line of the output of a program	
$N_{\text{True}}(P)$, $N_{\text{False}}(P)$	(5.1)	the results reported True and False by the program P
verdict	the last line `VERDICT: …` of the output, or, without it, "All printed results True"	the summary of a program
notebook check	a cell `check["name", test]` of 7+1.nb, or a check made by `frameChecks`	a test that returns True or False
N_{nb}, N_{printed}, N_{frame}	(5.2)	checks of the notebook; printed where they stand; made by `frameChecks`
`frameChecks`	a function of section 13.1 of 7+1.nb	nine checks of the frame matrices for each of four metrics
input, output cell	cells of the notebook	the Wolfram Language code and its results
message cell	a cell in which Mathematica reports a problem	must be 0
fresh kernel	a new Wolfram Language kernel with no earlier definitions	the evaluation starts from nothing
evaluation time	the duration of the evaluation of all cells	283.883 s for the evaluation that finished on 2026-10-28 08:57:47 PDT
rebuild	all documents written anew by `tools/make_docs.py` in a fresh clone	compared with the committed files
identical byte for byte	the same sequence of bytes	the comparison of Markdown and LaTeX files
same text and page count	equal `pdftotext` output and equal `pdfinfo` page count	the comparison of PDFs, which also record their creation time

$$N_{\text{True}}(P) = \#\{\ell :\ \ell \text{ ends in ": True" or in "(True)"}\}, \tag{5.1}$$
$$N_{\text{False}}(P) = \#\{\ell :\ \ell \text{ ends in ": False" or in "(False)"}\},$$
$$N_{\text{nb}} = N_{\text{printed}} + N_{\text{frame}}, \qquad N_{\text{frame}} = 9 \times 4 \qquad (479 = 443 + 36). \tag{5.2}$$

In (5.1) # counts the elements of a set, ℓ runs over the lines of the output of P, and spaces between the colon and the word, and after the word, are allowed.

6. The spreadsheet

symbol or term	definition	meaning
sheet	Summary, Commits, Documents, Fixes, Checks, Private files	the six tables of the spreadsheet
cell `B5`	column B, row 5	one field of a sheet
`Commits!E6`	the cell E6 of the sheet Commits	a reference to another sheet
`'Private files'!C2`	a sheet name with a space, quoted	
range `A2:A5`	the rectangle of cells from A2 to A5	
R, c, $v(c)$	a range, a cell, the value of the cell	
formula	a cell whose content begins with `=`	its value is computed from other cells
`SUM`, `COUNTA`, `COUNTIF`	(6.1)	the functions of the spreadsheet
`+`, `&`	(6.2)	addition; joining of texts
data file	tools/status_2026-10-28.json	every number of the spreadsheet
openpyxl	a Python library that writes `.xlsx` files	writes the spreadsheet with its formulas
LibreOffice (headless)	the office suite, run without a window	opens and saves the file, which computes and stores the formula values
stored value	the value of a formula saved in the file	what a viewer shows before computing anything

$$\mathrm{SUM}(R) = \sum_{c\in R,\ v(c)\text{ a number}} v(c), \qquad \mathrm{COUNTA}(R) = \#\{c \in R :\ v(c) \text{ not empty}\}, \tag{6.1}$$

$$\mathrm{COUNTIF}(R, x) = \#\{c \in R :\ v(c) = x\},$$

$$x + y = \text{the sum of } x \text{ and } y, \qquad x \,\&\, y = \text{the text of } x \text{ followed by the text of } y. \tag{6.2}$$

In `COUNTIF`, texts are compared without regard to case.

7. The time line of the clean-up

moment	definition	loose lines
first build	the first full build of the documents on 28 October 2026, before commit 0654dcd	40
after 0654dcd	the rebuild of a fresh clone of 0654dcd	12
now	the rebuild of a fresh clone of 102f399	0

8. Numbers and constants

symbol	value	meaning
G_F	1.1663788×10^{-5} GeV^{-2}	Fermi constant
v	246.2196402 GeV	electroweak scale, $(\sqrt{2}G_F)^{-1/2}$
M_{Pl}	2.435×10^{18} GeV	reduced Planck mass, $8\pi G_4 = M_{\mathrm{Pl}}^{-2}$
p_1, q_1, u_1	$\frac{1+2\sqrt{2}}{7}$, $u_1 p_1$, $-1 + \frac{1}{\sqrt{2}}$	Kasner constants, (9.3)
$10^{1/3}$	2.154	the stretch ratio of badness 1000

9. The symbols of the formulas of the rewritten paragraphs

"Row n" is the n-th row of the sheet Fixes, that is, the n-th rewritten paragraph. Each subsection gives the formula of a row as its document writes it, then its symbols; "(row n)" in the column "definition" means that the formula at the top of the subsection defines the symbol.

9.1 The Clifford products (row 2)

$$\Theta^K = i^{\sigma_r}\Gamma^{A_1}\cdots\Gamma^{A_r}, \qquad \sigma_r = \frac{r(r-1)}{2} \bmod 2, \qquad \Theta_K = \eta_{A_1A_1}\cdots\eta_{A_rA_r}\Theta^K.$$

symbol or term	definition	meaning
$\Gamma_A = \Gamma_A^{(1,7)}$	(9.1)	the eight real 16×16 Dirac matrices of 7+1.nb, $A = 0, \ldots, 7$
τ_i^O	real antisymmetric 8×8 matrices, (9.1)	the seven imaginary octonion units $e_1, \ldots, e_7$ as matrices
η_{AB}	$\mathrm{diag}(+1, -1, \ldots, -1)$	the squares of the octonion basis (section 1)
Γ^A	$\eta^{AA}\Gamma_A$ (no sum)	Dirac matrix with the index raised: $\Gamma^0 = \Gamma_0$, $\Gamma^i = -\Gamma_i$
$K = (A_1 < \cdots < A_r)$	an increasing list of indices	a multi-index; there are $\binom{8}{r}$ of rank r and 256 in all
$r = \lvert K\rvert$	$0, \ldots, 8$	the rank of the multi-index
σ_r	$\frac{r(r-1)}{2} \bmod 2$	0 or 1: the phase i^{σ_r}
Θ^K	$i^{\sigma_r}\Gamma^{A_1}\cdots\Gamma^{A_r}$	the 256 Clifford products ($\Theta^{()} = 1_{16}$)
Θ_K	$\eta_{A_1A_1}\cdots\eta_{A_rA_r}\Theta^K$	index lowered; $\Theta_K\Theta^K = 1_{16}$
i	$i^2 = -1$	the imaginary unit

$$\Gamma_0^{(1,7)} = \begin{pmatrix} 0 & 1_8 \\ 1_8 & 0 \end{pmatrix}, \qquad \Gamma_i^{(1,7)} = \begin{pmatrix} \tau_i^O & 0 \\ 0 & -\tau_i^O \end{pmatrix} \quad (i = 1, \ldots, 7), \tag{9.1}$$

$$\tau_i^O\tau_j^O + \tau_j^O\tau_i^O = -2\delta_{ij}1_8, \qquad \Gamma_A\Gamma_B + \Gamma_B\Gamma_A = 2\eta_{AB}1_{16}.$$

9.2 The Grassmann algebra (row 3)

$$\theta_K\theta_{K'} = \mathrm{sign}(\pi)\,\theta_{\mathrm{sort}(K\cup K')} \qquad (\text{zero if } K\cap K' \neq \emptyset),$$

$$\frac{\partial^L(\theta_{k_1}\cdots\theta_{k_r})}{\partial\theta_{k_j}} = (-1)^{j-1}\,\theta_{k_1}\cdots\widehat{\theta_{k_j}}\cdots\theta_{k_r}.$$

symbol or term	definition	meaning
θ_k	$\theta_k\theta_l = -\theta_l\theta_k$, so $\theta_k^2 = 0$	the anticommuting generators
K, K'	increasing lists of generator indices	index sets
θ_K	$\theta_{k_1}\cdots\theta_{k_r}$ for $K = (k_1 < \cdots < k_r)$	the basis monomials
π	the permutation that sorts K followed by K'	
$\mathrm{sign}(\pi)$	$+1$ for an even, -1 for an odd permutation	
$\mathrm{sort}(K\cup K')$	the union in increasing order	
$\partial^L/\partial\theta_{k_j}$	the left derivative	moves θ_{k_j} to the front, then removes it
$\widehat{\theta_{k_j}}$	the factor left out	

9.3 The xAct definitions (row 4)

```
CD = LC[g]; RiemannCD = Riemann[CD]; RicciCD = Ricci[CD];
RicciScalarCD = RicciScalar[CD]; EinsteinCD = Einstein[CD];
```

symbol or term	definition	meaning
xAct, xCoba, xTras	packages for the Wolfram Language	abstract and component tensor calculus
`g`	a component metric $g_{\mu\nu}$	the metric whose curvature is computed
`CD = LC[g]`	the Levi-Civita connection of `g`	after it, the covariant derivative `CD` is that of g
`RiemannCD`, `RicciCD`	(9.2)	the Riemann and Ricci tensors of g
`RicciScalarCD`, `EinsteinCD`	(9.2)	the scalar curvature and the Einstein tensor of g
$\{^{\lambda}_{\mu\nu}\}$	(9.2)	Christoffel symbols

$$\{^{\lambda}_{\mu\nu}\} = \tfrac{1}{2}\, g^{\lambda\rho}\big(\partial_\mu g_{\rho\nu} + \partial_\nu g_{\rho\mu} - \partial_\rho g_{\mu\nu}\big), \qquad R^{\rho}{}_{\sigma\mu\nu} = \partial_\mu\{^{\rho}_{\sigma\nu}\} - \partial_\nu\{^{\rho}_{\sigma\mu}\} + \{^{\rho}_{\mu\lambda}\}\{^{\lambda}_{\sigma\nu}\} - \{^{\rho}_{\nu\lambda}\}\{^{\lambda}_{\sigma\mu}\},$$
$$R_{\sigma\nu} = R^{\rho}{}_{\sigma\rho\nu}, \qquad R = g^{\sigma\nu}R_{\sigma\nu}, \qquad G_{\mu\nu} = R_{\mu\nu} - \tfrac{1}{2}g_{\mu\nu}R. \tag{9.2}$$

These are the conventions of 7+1.nb (in the Landau–Lifshitz convention for the Riemann tensor); $g^{\lambda\rho}$ is the inverse metric and $\partial_\mu = \partial/\partial x_\mu$.

9.4 The mixing angle in matter (row 5)

$$\frac{\Delta m^2}{4E}\begin{pmatrix} -\cos 2\theta & \sin 2\theta \\ \sin 2\theta & \cos 2\theta \end{pmatrix} + \mathrm{diag}(V,0) \quad\Longrightarrow\quad \tan 2\theta_m = \frac{\sin 2\theta}{\cos 2\theta - 2EV/\Delta m^2}.$$

symbol or term	definition	meaning
Δm^2	$m_2^2 - m_1^2$	difference of the squared masses of two neutrinos
E	energy	the energy of the neutrino
θ	the mixing angle in vacuum	
V	$\sqrt{2}G_F n_e$	the potential of the electron neutrino in matter of electron density n_e
θ_m	(row 5)	the mixing angle in matter
$\mathrm{diag}(V,0)$	$\left(\begin{smallmatrix} V & 0 \\ 0 & 0 \end{smallmatrix}\right)$	the matter term of the Hamiltonian

9.5 The mode equation of pair creation (row 6)

$$i\,\frac{dc}{d\tau} = \omega\,(\cos\theta\,\sigma_1 + \sin\theta\,\sigma_2)\,c, \qquad \omega = K\sqrt{\tau^{-2p_1} + \tau^{-2q_1}}, \qquad \tan\theta = \tau^{p_1 - q_1}.$$

symbol or term	definition	meaning
$c(\tau)$	a column of two complex numbers	the amplitude of a mode
τ	$x_0 - x_s$	time since the singularity at $x_0 = x_s$
K	a positive number	the scaling variable of the mode (not a multi-index here)
ω	$K\sqrt{\tau^{-2p_1} + \tau^{-2q_1}}$	the frequency of the mode
θ	$\tan\theta = \tau^{p_1 - q_1}$	the angle of the mode Hamiltonian (not the angle of §9.4)
σ_1, σ_2	(9.3)	Pauli matrices
p_1, q_1, u_1	(9.3)	the Kasner exponents of ordinary and auxiliary space in the earliest regime

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad u_1 = -1 + \frac{1}{\sqrt{2}}, \qquad p_1 = \frac{1+2\sqrt{2}}{7}, \qquad q_1 = u_1 p_1,$$

$$3p_1 + 4q_1 = 1, \qquad 3p_1^2 + 4q_1^2 = 1 \qquad (p_1 \approx 0.5469,\ q_1 \approx -0.1602). \tag{9.3}$$

9.6 The minisuperspace action (rows 7 and 8)

$$S = -\frac{1}{2\kappa} \int n\, a^3 b^4 \sum_k \alpha_k L_k \, dx_0 - \int nW \, dx_0,$$

$$\mathrm{EL}_n(na^3b^4L_k) = -2a^3b^4 E^{(k)0}{}_0, \qquad \mathrm{EL}_a(na^3b^4L_k) = -6a^2b^4 E^{(k)1}{}_1,$$

$$\mathrm{EL}_b(na^3b^4L_k) = -8a^3b^3 E^{(k)4}{}_4.$$

symbol or term	definition	meaning
n, a, b	(9.4)	lapse; scale factors of ordinary and of auxiliary space; functions of x_0
S	(row 7)	the action restricted to the metrics (9.4)
κ	$8\pi G_8 = M_8^{-6}$	the eight-dimensional gravitational coupling
α_k	$\alpha_0 = 2\Lambda$, $\alpha_1 = 1$	the Lovelock couplings (Λ the cosmological constant)
L_k	(9.5)	the Lovelock densities: $L_0 = 1$, $L_1 = R$
$\delta^{\mu_1\cdots\mu_p}_{\nu_1\cdots\nu_p}$	(9.5)	the generalized Kronecker delta
$E^{(k)\mu}{}_{\nu}$	(9.5)	the Lovelock tensors: $E^{(0)} = -\delta/2$, $E^{(1)} = G$ (the Einstein tensor)
$W(a,b)$	$a^3b^4\varepsilon$	the comoving energy of the matter (ε its energy density)
EL_q	(9.6)	the Euler–Lagrange expression for the variable $q = n, a, b$

$$ds^2 = n^2 dx_0^2 - a^2\big(dx_1^2 + dx_2^2 + dx_3^2\big) - b^2\big(dx_4^2 + \cdots + dx_7^2\big), \tag{9.4}$$

$$L_k = \frac{1}{2^k} \delta^{\alpha_1\beta_1\cdots\alpha_k\beta_k}_{\gamma_1\delta_1\cdots\gamma_k\delta_k} R^{\gamma_1\delta_1}{}_{\alpha_1\beta_1} \cdots R^{\gamma_k\delta_k}{}_{\alpha_k\beta_k},$$

$$E^{(k)\mu}{}_{\nu} = -\frac{1}{2^{k+1}} \delta^{\mu\alpha_1\beta_1\cdots\alpha_k\beta_k}_{\nu\gamma_1\delta_1\cdots\gamma_k\delta_k} R^{\gamma_1\delta_1}{}_{\alpha_1\beta_1} \cdots R^{\gamma_k\delta_k}{}_{\alpha_k\beta_k},$$

$$\delta^{\mu_1\cdots\mu_p}_{\nu_1\cdots\nu_p} = \det\left(\delta^{\mu_i}_{\nu_j}\right)_{i,j=1}^{p}, \tag{9.5}$$

$$\mathrm{EL}_q(F) = \frac{\partial F}{\partial q} - \frac{d}{dx_0}\frac{\partial F}{\partial \dot q} + \frac{d^2}{dx_0^2}\frac{\partial F}{\partial \ddot q} \qquad (\dot q = dq/dx_0). \tag{9.6}$$

The Greek indices $\alpha_j, \beta_j, \gamma_j, \delta_j$ of (9.5) are summation indices over $0, \ldots, 7$, not the couplings α_k; $R^{\gamma\delta}{}_{\alpha\beta}$ is the Riemann tensor of (9.2) with two indices raised.

9.7 The comoving energy of one massless particle (row 8)

$$W = \omega = \sqrt{k_a^2/a^2 + k_b^2/b^2}, \qquad \partial_a W = -\frac{k_a^2}{a^3\omega}, \qquad \partial_b W = -\frac{k_b^2}{b^3\omega}.$$

symbol or term	definition	meaning
k_a, k_b	constants	the comoving momenta of the particle along ordinary and along auxiliary space
ω	$\sqrt{k_a^2/a^2 + k_b^2/b^2}$	its energy (not the frequency of §9.5)

symbol or term	definition	meaning
∂_a, ∂_b	$\partial/\partial a$, $\partial/\partial b$	partial derivatives with respect to the scale factors

9.8 The Kaluza–Klein level and the radion (row 9)

$$M_E(\beta) = e^{-2\beta}\sqrt{m^2 + \mu_n^2 e^{-2\beta}}, \qquad g_n = \Big|\frac{dM_E}{d\phi}\Big|_{\beta=0} = \frac{2M_n + \mu_n^2/M_n}{\sqrt{12}\,M_{\rm Pl}}.$$

symbol or term	definition	meaning
β	$\ln(b/b_0)$	the radion: the logarithm of the size b of the auxiliary space (b_0 its stabilized value, at the minimum of the stabilizing potential)
ϕ	$\sqrt{12}\,M_{\rm Pl}\beta$	the canonical radion
$M_{\rm Pl}$	2.435×10^{18} GeV	the reduced Planck mass
m	a mass	the mass of dirac16complex
μ_n	$2\pi\lvert\vec{n}\rvert/\ell_\perp$ on the auxiliary torus	the Kaluza–Klein mass of the level n ($\ell_\perp$ the size of the torus, $\vec{n} \in \mathbb{Z}^4$)
$M_E(\beta)$	(row 9)	the mass of the level in the Einstein frame
M_n	$M_E(0) = \sqrt{m^2 + \mu_n^2}$	the mass of the level today
g_n	$\lvert dM_E/d\phi\rvert$ at $\beta = 0$	the coupling of the level to the radion

$$\frac{dM_E}{d\beta}\Big|_{\beta=0} = -2M_n - \frac{\mu_n^2}{M_n}, \qquad \frac{d}{d\phi} = \frac{1}{\sqrt{12}\,M_{\rm Pl}}\frac{d}{d\beta} \qquad \Longrightarrow \qquad g_n = \frac{2M_n + \mu_n^2/M_n}{\sqrt{12}\,M_{\rm Pl}}.$$

9.9 The five momenta (row 10)

$$\vec{p} \in \big\{ \pm(2,2,4,0,0,0,0),\ \pm(0,0,0,2,2,0,0),\ \vec{0}\big\} \qquad (E = 5,5,3,3,1).$$

symbol or term	definition	meaning
$\vec{p}$	$(p^1, \ldots, p^7)$	the momentum of a plane wave
E	$\sqrt{m^2 + \lvert\vec{p}\rvert^2}$	its energy; with $m = 1$: 5, 5, 3, 3, 1
$\vec{x}$, $\vec{y}$	points of space $(x_1, \ldots, x_7)$	the arguments of the field in the sum

9.10 The orthogonal matrix of the spinors (row 11)

$$U = \frac{1}{\sqrt{2E}}\big(u_1(\vec{p}), \ldots, u_8(\vec{p}), v_1(-\vec{p}), \ldots, v_8(-\vec{p})\big) \qquad \text{satisfies} \qquad U^{\mathsf{T}} h(\vec{p})\, U = \mathrm{diag}(E1_8, -E1_8).$$

symbol or term	definition	meaning
e_s	the unit columns of length 8, $s = 1, \ldots, 8$	
$M(\vec{p})$	$m1_8 + p^i \tau_i^O$	the real 8×8 block; $MM^{\mathsf{T}} = E^2 1_8$
α_i	$\Gamma^0\Gamma^i$	Dirac's α matrices; real symmetric
$h(\vec{p})$	$p^i\alpha_i + m\Gamma^0$	the Dirac Hamiltonian of a plane wave; real symmetric
$u_s(\vec{p})$	(9.7)	the real spinors of positive energy
$v_s(\vec{p})$	(9.7)	the real spinors of negative energy
U	(row 11)	the real orthogonal 16×16 matrix of the spinors
$E1_8$	E times the identity	

$$u_s(\vec{p}) = \frac{1}{\sqrt{E}} \begin{pmatrix} E\, e_s \\ M(\vec{p})^{\mathsf{T}} e_s \end{pmatrix}, \qquad h(\vec{p})\, u_s = E\, u_s, \qquad h(-\vec{p})\, v_s = -E\, v_s, \qquad u_s^{\mathsf{T}} u_r = v_s^{\mathsf{T}} v_r = 2E\, \delta_{sr}. \quad (9.7)$$

9.11 Completeness and the Fierz identity (row 14)

$$\sum_K (\Theta_K)_{cd} (\Theta^K)_{ab} = 16\, \delta_{ad} \delta_{bc},$$

$$\sum_{K \in r} (a \Theta_K b)(c \Theta^K d) = \sum_J C_{rJ}\, (a \Theta_J d)(c \Theta^J b), \qquad C_{rJ} = \frac{1}{256} \sum_{K \in r} \mathrm{tr}\big(\Theta_K \Theta_J \Theta^K \Theta^J\big).$$

symbol or term	definition	meaning
$(\Theta_K)_{cd}$	the entry in row c, column d	$a, b, c, d = 1, \ldots, 16$ as subscripts are spinor indices
a, b, c, d	commuting columns of 16 numbers	spinors, in the second display
(aMb)	$\sum_{i,j} a_i M_{ij} b_j$	the bilinear of a 16×16 matrix M
$K \in r$	the multi-indices of rank r	
J	a multi-index (any of the 256)	
C_{rJ}	(row 14)	the Fierz coefficients
tr	the trace, the sum of the diagonal entries	

9.12 The notation of sections 7–10 of 7+1.nb (row 15)

$$\Gamma_A = \Gamma_A^{(1,7)}, \qquad \Gamma^A = \eta^{AA}_{(1,7)} \Gamma_A, \qquad \Gamma^0 = \Gamma_0, \qquad \chi = \chi^{(1,7)} = i\Gamma_8^{(1,7)},$$
$$\bar{\Psi} = \Psi^\dagger \Gamma^0, \qquad \vec{x} = (x_1, \ldots, x_7), \qquad \vec{p} = (p^1, \ldots, p^7).$$

symbol or term	definition	meaning
Ψ	a column of 16 anticommuting operators	the quantized field dirac16complex
$\Psi^\dagger$	its conjugate transpose	
$\bar{\Psi}$	$\Psi^\dagger \Gamma^0$	the Dirac adjoint
$\eta_{(1,7)}$	$\mathrm{diag}(+1, -1, \ldots, -1)$	the same matrix as η
$\Gamma_8^{(1,7)}$	(9.8)	the volume element, $(\Gamma_8^{(1,7)})^2 = -1_{16}$
$\chi = \chi^{(1,7)}$	$i\Gamma_8^{(1,7)}$	the (complex) chirality matrix, $\chi^2 = 1_{16}$
$\vec{x}$, $\vec{p}$	$(x_1, \ldots, x_7)$, $(p^1, \ldots, p^7)$	space point and momentum

$$\Gamma_8^{(1,7)} = \Gamma_0^{(1,7)} \Gamma_1^{(1,7)} \cdots \Gamma_7^{(1,7)} = \begin{pmatrix} 0 & -1_8 \\ 1_8 & 0 \end{pmatrix}. \quad (9.8)$$

9.13 The electroweak scale (row 16)

$$v = (\sqrt{2} G_F)^{-1/2} = 246.2196402 \text{ GeV}, \qquad m_W = \frac{gv}{2}, \qquad m_Z = \frac{\sqrt{g^2 + g'^2}\, v}{2}, \qquad m_H^2 = 2\lambda v^2.$$

symbol or term	definition	meaning
G_F	1.1663788×10^{-5} GeV^{-2}	the Fermi constant

symbol or term	definition	meaning
v	$(\sqrt{2}G_F)^{-1/2} = 246.2196402$ GeV	the electroweak scale (the vacuum value of the Higgs field)
g, g'	dimensionless	the gauge couplings of $SU(2)_L$ and $U(1)_Y$
λ	dimensionless	the quartic coupling of the Higgs field
m_W, m_Z, m_H	(row 16)	the masses of the W, Z and Higgs bosons at tree level (lowest order)

9.14 The matrix notation (row 17)

$$1_n = (\delta_{ij})_{i,j=1}^{n}, \qquad (M^\dagger)_{ij} = \overline{M_{ji}}, \qquad (M^*)_{ij} = \overline{M_{ij}},$$
$$(A \otimes B)_{(i-1)p+k,\,(j-1)q+l} = A_{ij} B_{kl} \quad (B \text{ of size } p \times q).$$

symbol or term	definition	meaning
1_n	(δ_{ij})	the $n \times n$ identity
M^T	$(M^\mathsf{T})_{ij} = M_{ji}$	the transpose
$M^\dagger$	$(M^\dagger)_{ij} = \overline{M_{ji}}$	the conjugate transpose
M^*	$(M^*)_{ij} = \overline{M_{ij}}$	the complex conjugate
$\overline{z}$	the complex conjugate of the number z	
$A \otimes B$	(row 17)	the Kronecker product; in Mathematica `KroneckerProduct[A, B]`

10. File and program names

name	what it is
`7plus1_status_2026-10-28.xlsx`	the spreadsheet: the sheets Summary, Commits, Documents, Fixes, Checks, Private files
`tools/status_2026-10-28.json`	the data file: every number of the spreadsheet
`tools/make_status_xlsx.py`	writes the spreadsheet from the data file
`tools/make_docs.py`	writes every document (Markdown, LaTeX, PDF) and runs the twelve check programs
`tools/doc_status.py`, `tools/doc_status_symbols.py`, `tools/doc_status_provenance.py`, `tools/doc_status_common.py`	write the document on the state of the repository, this dictionary and the provenance record
`build/docs/runs.json`	the outputs of the twelve programs in the last build (build/ is not published)
`tools/eval_log_7p1.txt`	the record of the evaluation of 7+1.nb
`tools/verify_7p1_nb.py`	parses the evaluated 7+1.nb and compares it with the Python programs and the backups
`tools/ref44.py`, `tools/ref71.py`, `tools/extra71.py`	rebuild 4+4.nb and the objects of 7+1.nb; further identities
`tools/ref_dirac16.py`, `tools/ref_gravity.py`, `tools/ref_ks_gravity.py`, `tools/ref_sm.py`	independent checks of sections 7–9, 11–14, 15–16 and 17–19 of 7+1.nb
`tools/ref_coupled.py`, `tools/ref_dark.py`, `tools/ref_torus_brane.py`, `tools/ref_radion.py`	independent checks of sections 20–23, 24, 25 and 26 of 7+1.nb
`tools/doc_stage3_common.py`	shared data of the documents of sections 11–26; its SHA-256 is recorded in the provenance files
7+1.nb	the notebook of the project (Wolfram Language 15.0.1), evaluated
Einstein-Lovelock-7+1-Nash.nb	the author's reference notebook for the Lovelock tensors with xAct

11. Symbols that look alike

symbol	**is not**
Γ_A, Γ^A (Dirac matrices, one index)	$\{^{\lambda}_{\mu\nu}\}$ (Christoffel symbols, never written Γ here)
K (multi-index, §9.1, §9.11)	K (the scaling variable of a mode, §9.5)
θ (mixing angle in vacuum, §9.4)	θ (angle of the mode Hamiltonian, §9.5); θ_k, θ_K (Grassmann generators, §9.2); θ_m (mixing angle in matter, §9.4)
σ_r (0 or 1, §9.1)	σ_1, σ_2 (Pauli matrices, §9.5)
ω (frequency of a mode, §9.5)	ω (energy of one massless particle, §9.7)
E (energy)	$E^{(k)\mu}{}_{\nu}$ (Lovelock tensors, §9.6); $E1_8$ (E times the identity)
a, b (scale factors, §9.6)	a, b, c, d (spinors or spinor indices, §9.11)
$c(\tau)$ (mode amplitude, §9.5)	c (a spinor or a spinor index, §9.11); c (a cell of the spreadsheet, section 6)
β (radion, §9.8)	α_k (Lovelock couplings) and the summation indices α_j, β_j of (9.5)
g (a metric, §9.3)	g, g' (gauge couplings, §9.13); g_n (coupling of a level to the radion, §9.8)
R (scalar curvature, §9.3)	R (lines removed, section 2); R (a range of cells, section 6)
s (stretchability, section 3)	s (spinor label $s = 1, \ldots, 8$, §9.10)
ℓ (a line of output, section 5)	$\ell_\perp$ (the size of the auxiliary torus, §9.8)
η, $\eta_{(1,7)}$ (the same matrix)	$\eta_{71} = -\eta$ (the sign convention of 4+4.nb, section 1)